Gradient Descent – Complete Explanation

1. What is Gradient?
Gradient is the slope of the loss function with respect to a parameter (weight). It tells how much the loss will change if the weight changes slightly.

2. Meaning of Gradient Value
- Positive Gradient → Decrease weight
- Negative Gradient → Increase weight
- Zero Gradient → Minimum loss reached

3. Gradient Descent Update Rule
New Weight = Old Weight – (Learning Rate × Gradient)

4. Loss Function Example (MSE)
Loss = $(\text{Predicted} - \text{Actual})^2$

5. Gradient Formula (Linear Model)
y = wx
Gradient = 2x(y_pred – y_actual)

6. Code Example (Manual Gradient)
x = 2, y = 10, w = 3
y_pred = w × x
loss = $(\text{y\_pred} - \text{y})^2$
gradient = 2 × x × (y_pred – y)

7. Weight Update Example
w_new = w – learning_rate × gradient

8. Real-World Example
In house price prediction, if the predicted price is higher than actual, the gradient becomes positive and the weight is reduced to lower the prediction.

9. Automatic Gradient (PyTorch)
Frameworks compute gradients automatically using backpropagation.

10. Key Points
- Gradient shows direction of error change
- Learning rate controls step size
- Gradient descent minimizes loss step-by-step